

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

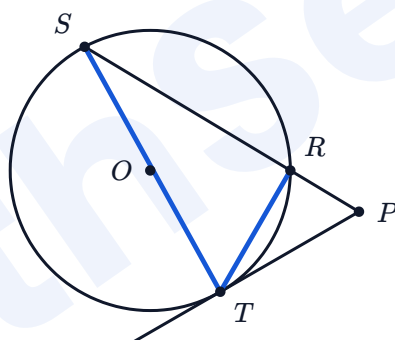
Chapter 18: Tangents and Intersecting Chords

Previous year board questions, topic-wise: 6 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Tangent and secant from an external point

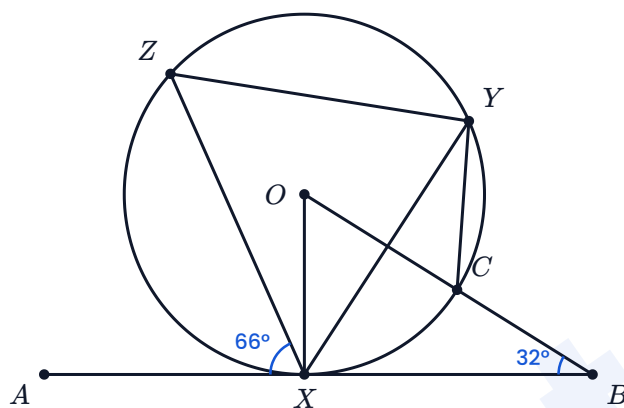
- PT is a tangent to a circle at T , and a line through P cuts the circle at R and S , with R nearer to P . (a) Prove that $\triangle PTR \sim \triangle PST$. (b) Prove that $PT^2 = PR \times PS$. (c) If $PR = 4$ cm and $PS = 16$ cm, find PT . **ICSE 2025, 3 marks**



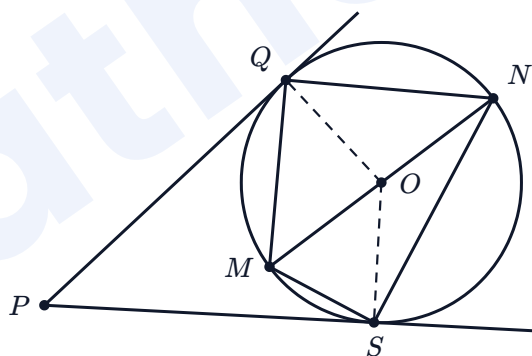
- A circle has centre O and radius 9 cm. The diameter AB is produced to a point P , and PQ is a tangent from P . If $PA = 24$ cm, find PQ . **ICSE 2024, 3 marks**

2 The alternate segment theorem

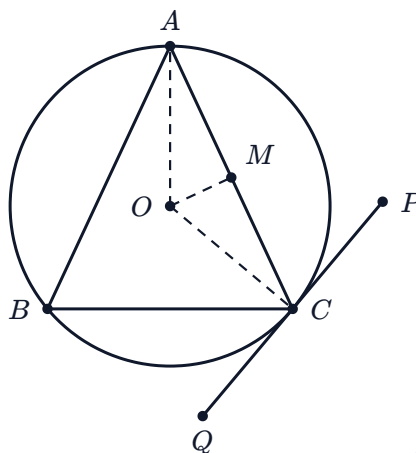
3. X, Y, Z and C lie on a circle with centre O . AB is the tangent at X , OB cuts the circle at C , and $ZY = XY$. Given $\angle OBX = 32^\circ$ and $\angle AXZ = 66^\circ$, find (a) $\angle BOX$ (b) $\angle CYX$ (c) $\angle ZYX$ (d) $\angle OXY$. **ICSE 2025, 4 marks**



4. PR and PT are tangents from P to a circle with centre O , touching it at Q and S . MN is a diameter, with M on the arc QS nearer to P . Given $\angle PQM = 42^\circ$ and $\angle PSM = 25^\circ$, find (a) $\angle OQM$ (b) $\angle QNS$ (c) $\angle QOS$ (d) $\angle QMS$. **ICSE 2024, 4 marks**



5. An isosceles triangle ABC with $AB = AC$ is inscribed in a circle with centre O . PQ is the tangent at C , with Q on the side of BC away from A . OM is perpendicular to the chord AC and $\angle COM = 65^\circ$. Find (a) $\angle ABC$ (b) $\angle BAC$ (c) $\angle BCQ$. **ICSE 2024, 3 marks**



3 Tangent properties

6. PS and PT are tangents to a circle from P , with $\angle SPT = 80^\circ$. Q is a point on the circle with $SQ \parallel PT$. Find $\angle QST$. (a) 140° (b) 90° (c) 80° (d) 50° **ICSE 2024, 1 mark**

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